Airline Operations Performance Analysis

BAN 650 - Data Visualization for Managers

Group Capstone Project | Spring 2026

Project focus	Managerial analysis of airline operations, delays, cancellations, trends, and route performance
Dataset	Class Project Airline Flight Data
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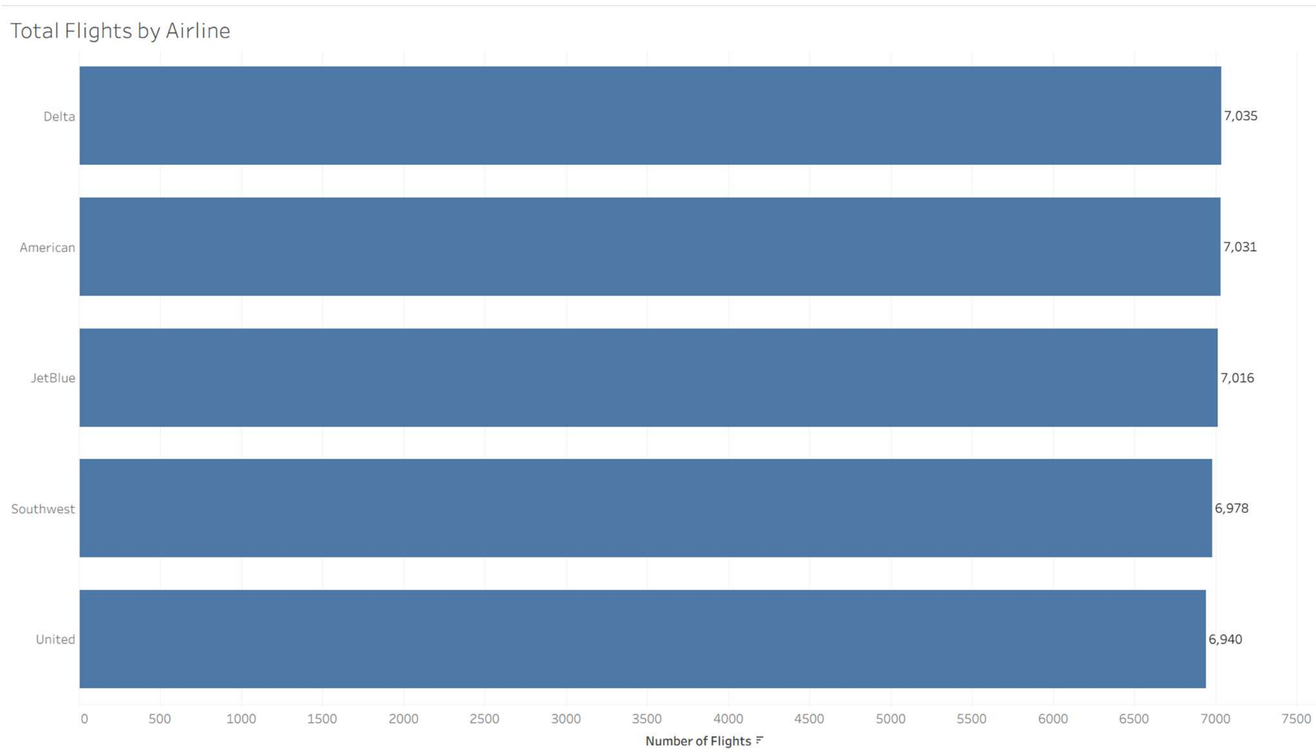
Project Overview

This project uses the Class Project Airline Flight dataset to analyze airline operational performance through data visualization. The goal of the report is to help managers better understand patterns in flight activity, delays, cancellations, and other operational trends by using clear and relevant visuals. The project begins with foundational analysis to establish overall performance, then moves into more advanced visualizations that provide deeper context through time-based, geographic, and multivariable views. The report concludes by connecting those findings to managerial decision-making, while also evaluating the strengths, limitations, and ethical considerations of the visualizations used.

1. Foundational Analysis: Flight Volume

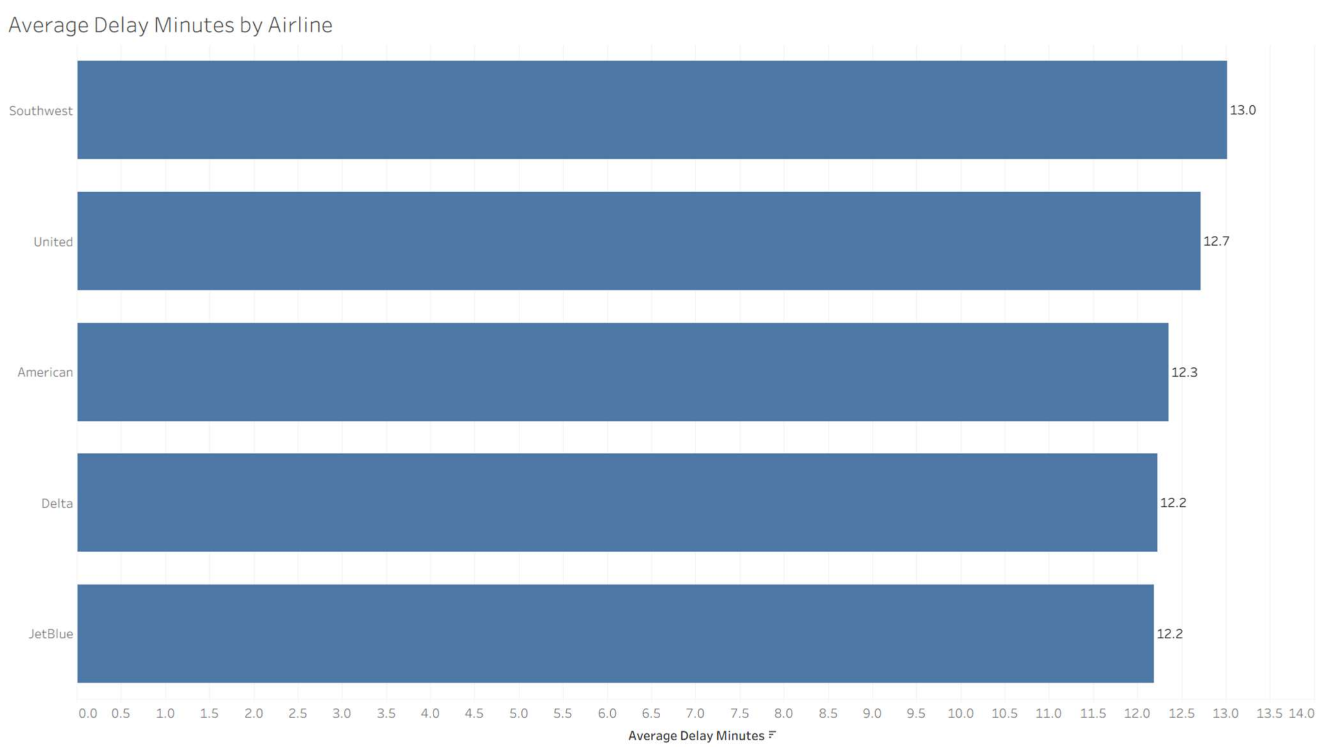
Figure 1. Total Flight Volume by Airline



The chart answers the initial business question of the distribution of the total flight volume among the airlines in the dataset. The best fit in this case is a bar chart since it enables managers to compare the airline volume easily and in a short time across categories. The most significant trend is that the number of flights is almost the same for all five airlines, with Delta having the largest number and United the minimum, but the difference is not that significant. The implication of that contrast is that the dataset represents a relatively balanced operating scale, as opposed to one airline controlling the overall activity. It is on this basis that management must have before considering delays, cancellations or route performance since raw operational problems may appear more serious when an airline merely has more flights. This visual provides a reasonable starting point to compare performance outcomes with the rest of the report since it establishes that the flight volume is relatively even.

2. Foundational Analysis: Average Delay Comparison

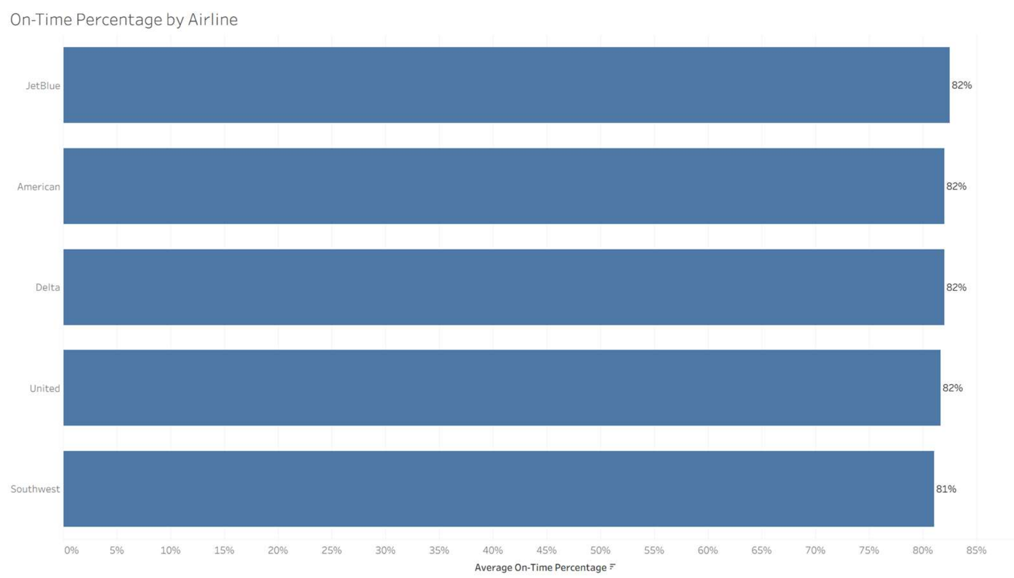
Figure 2. Average Delay Minutes by Airline



This chart is used to compare the average delay in minutes of the airlines in the dataset where average delay is used as a performance measure. The trend indicates that Southwest has the lowest relative delay performance, and JetBlue has the best, with the other airlines falling very close to each other. A significant difference with the flight-volume number is that the airline activity levels were relatively balanced, but the delay performance is not ranked equally, implying that the differences in delay are not merely a consequence of volume. Averages would be beneficial in this case since they would provide a more equitable comparison among airlines as opposed to raw delay totals would. Meanwhile, average values may mask variation, i.e. an airline with a similar average may nonetheless have more extreme delay spikes on some routes or days. To management, this observation is a tool to see what airlines can be subjected to closer operational scrutiny in the evaluation of reliability and on-time performance.

3. Foundational Metric Extension

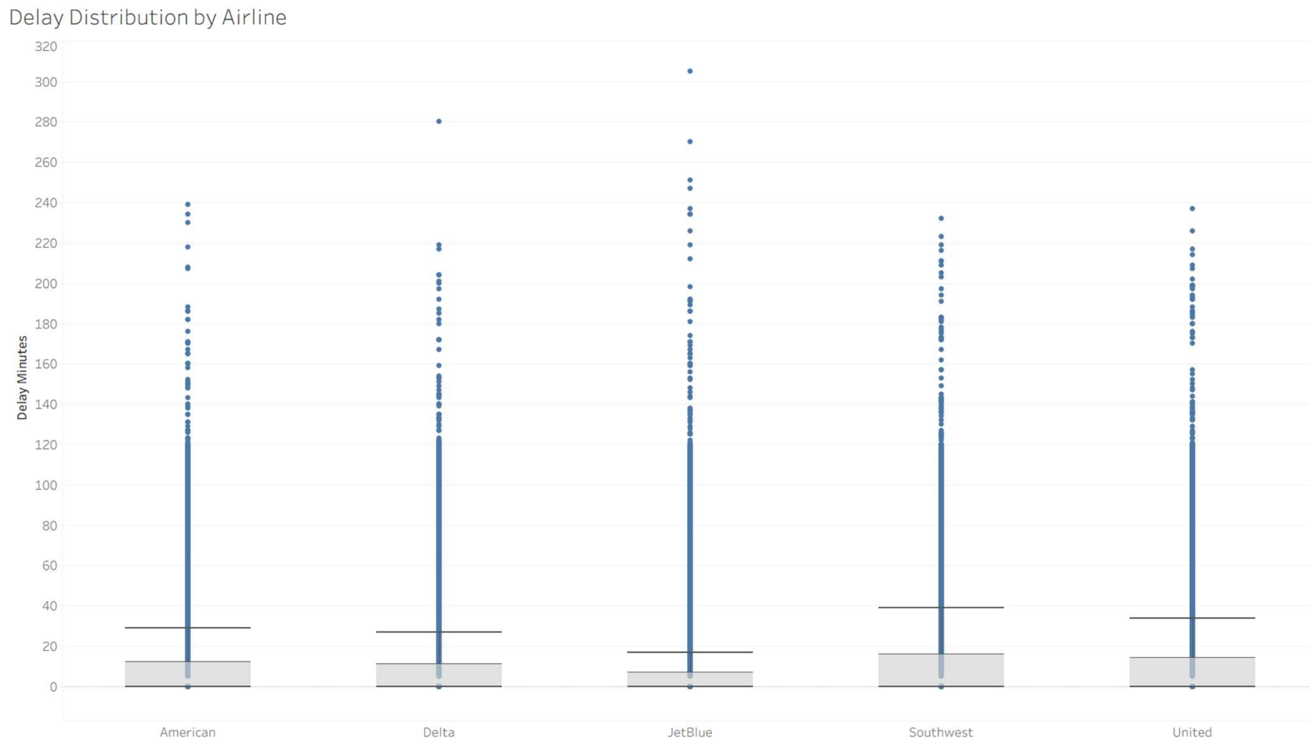
Figure 3. Additional Foundational Metric



The average on-time percentage by airline visualization provides a direct snapshot of operational reliability and overall customer experience for each airline under review. Figures one and two are focused on visualizing the scale of operations and the duration of flight delays; this chart above introduces a standardized ratio of the successes to failures for the airline carrier’s punctuality. The bar chart reveals a high degree of industry performance with four of the five major airlines, JetBlue, American, Delta, and United being tied at an 82% on-time performance. Southwest is seen as the clear trailer of the group with an 81% on-time performance rate. Small differences can be seen within the four airlines that scored an 82% rate, but the clustering suggests that on-time performance has become a standard in the industry even when minor deviations are visible. Management can see these results as a signal that they must prioritize marginal strategies to improve their on-time performance so that they do not fall behind the industry average, as Southwest has, in such a highly competitive market.

4. Distribution View

Figure 4. Distribution View of Delay Performance



The distribution view highlights the variation in delay performance across airlines rather than focusing only on average values. The chart reveals a right-skewed distribution, where most flights experience relatively low delays, but there are frequent extreme outliers representing significantly higher delay times. These outliers indicate that while average delay may appear similar across airlines, the consistency of performance differs meaningfully. Some airlines exhibit tighter clustering of delays, suggesting more stable operations, while others show wider spread and more extreme values, indicating greater variability and risk. This figure is important because it builds on the earlier average-based comparisons by showing that similar averages can mask underlying instability in performance. From a managerial perspective, this variation represents operational risk, as inconsistent delays can negatively impact reliability and customer satisfaction. Relying on averages alone would therefore be misleading, as it does not capture the frequency or severity of extreme delay events.

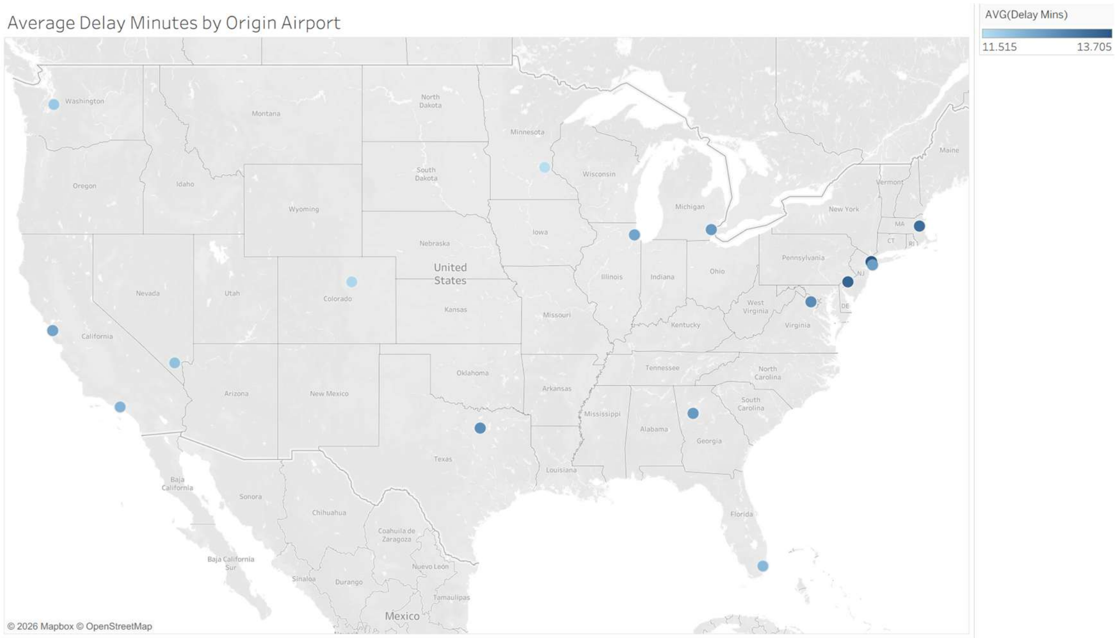
5. Temporal Analysis

Figure 5. Time-Based Trend in Airline Performance

The time-based trend chart analyzes average delay minutes by month, which is an appropriate interval because it allows management to evaluate performance patterns over a full operating year while capturing short-term fluctuations. The visual shows a moderate fluctuation pattern rather than a consistent upward or downward trend, with noticeable peaks in the summer and late fall months and a significant dip around August. This suggests that airline delay performance is not stable over time and may be influenced by seasonal factors such as travel demand, weather conditions, or operational strain during peak periods. From a managerial perspective, this pattern is important because it highlights specific time periods where performance weakens, rather than indicating a system-wide issue across all months. The variation appears to be more than random noise, as the repeated increases and decreases suggest a seasonal or cyclical pattern rather than isolated spikes. This implies that management should focus on targeted operational improvements during high-delay months, such as adjusting scheduling buffers or increasing staffing during peak travel periods. At the same time, since delays return to lower levels in other months, the pattern does not indicate a permanent decline in performance but rather a recurring operational challenge that can be managed with better planning.

6. Geospatial Analysis

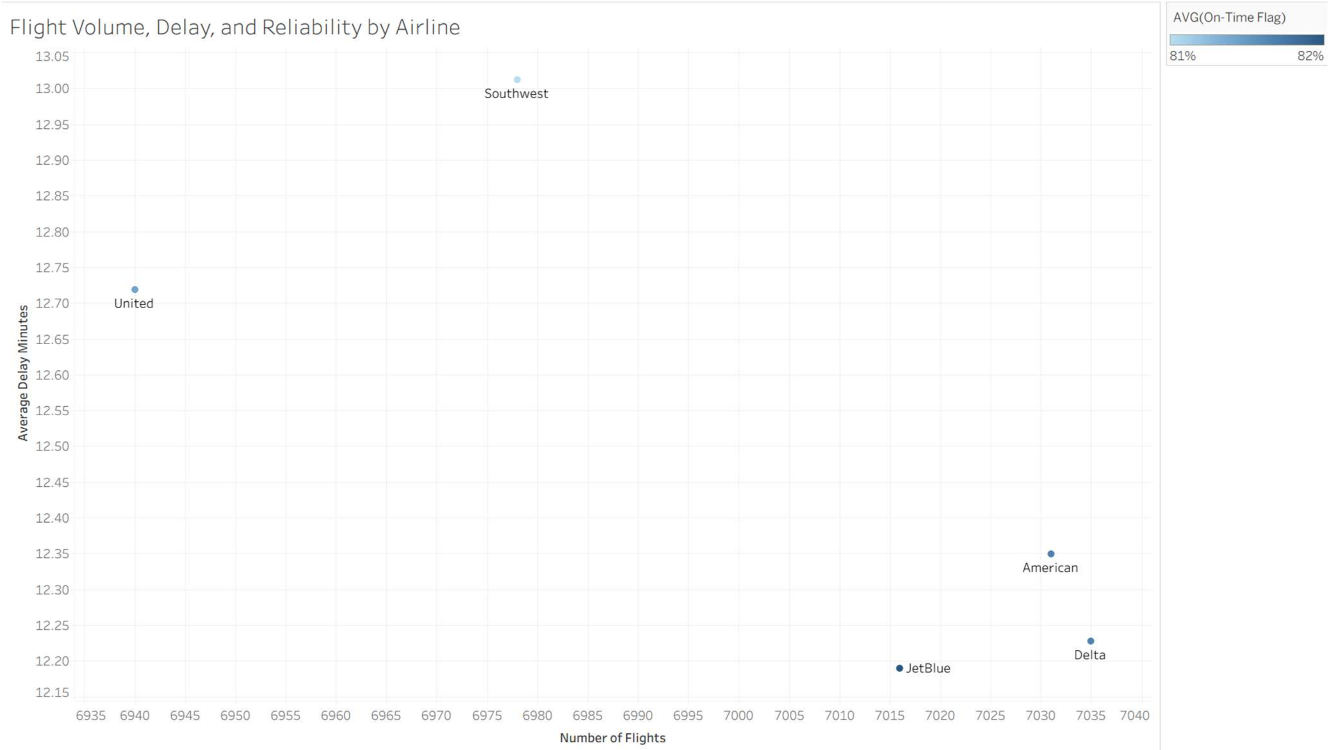
Figure 6. Geographic Pattern of Airline Activity or Delay Pressure



The map answers whether average flight delay varies systematically with airport geography, and a map is the appropriate form here because a bar chart would preserve the ranking but discard the spatial relationships between airports. The key insight is that delay is remarkably consistent across the network: the 16 airports span only 2.2 minutes (11.5 at MSP to 13.7 at LGA), with a mild east-leaning gradient where the three highest averages (LGA, PHL, BOS) all sit on the Northeast corridor. For management, this suggests that if any operational review is warranted it should start with Northeast corridor operations, but because the variation is so narrow, system-wide levers like scheduling buffers or turnaround procedures are likely to produce larger gains than airport-specific programs. The main limitation, and the central ethical concern, is that Tableau's auto-scaled color range stretches a 2.2-minute spread across the full palette, which makes the geographic pattern look far more dramatic than the underlying data supports and invites automation bias if a manager accepts the visual ranking without checking magnitudes. The map also lacks cause-of-delay context (weather, air traffic control, mechanical) and does not encode flight volume, so it should be read as a starting point for investigation rather than a performance scorecard.

7. Multivariate or Grouped Analysis

Figure 7. Multivariate or Grouped Comparison



This plot combines the number of flights, average delay minutes, and average on-time percentage to evaluate how operational scale influences service reliability. By combining these metrics, it answers the question of whether higher flight volumes lead to bad performance or longer wait times. A cluster is visible at the bottom right, where Delta, American, and JetBlue maintain a high volume of flights with minimal delays, while Southwest stands out as having the highest average delay minutes. This visualization offers more insight than a simple table by showing the trade-offs between capacity and efficiency across a very competitive industry. Managers should first notice the performance gap between the cluster of Delta, American, JetBlue, and Southwest. This view allows for the ranking of airline health that single-metric charts would obscure, showing that high volume does not always necessitate high delays. However, analysts and managers should be careful when interpreting the x-axis, as it is highly compressed. The actual difference between the lowest and highest flight volume shown is less than 2%.

8. Final Narrative, Evaluation, and Close

The visuals in Figures 1–3 establish the baseline for understanding the airline's performance in the dataset. Since the number of flights is almost the same for all airlines, flight volume is relatively balanced, and no carrier is dominating the operational scale. With that being said, Figure 2 does show that delay performance doesn't follow the same pattern. Southwest, in Figure 2, shows weaker average delay outcomes while JetBlue performs comparatively better. Figure 3 backs up this contrast by showing small, meaningful differences in the on-time percentage, even though the airlines operate at a similar scale. Altogether, our findings demonstrate that performance differences exist independently of airline size, meaning managers can't just instantly assume that higher volume alone explains the weaker reliability.

Figures 4 and 5 help us see why averages alone aren't enough for managerial decision-making. The distribution view shows a right-skewed distribution with frequent extreme outliers, indicating that some of the airlines experience more volatile delay patterns, even when their averages look similar to one another. This shows operational risk that would be invisible in a simple mean comparison. Figure 5 adds a temporal dimension, showing that delay performance fluctuates across months. Peaks in the summer and fall tend to contrast with dips in August, telling us that performance varies across airlines, flights, and time. Wrapping up for these Figures, managers have to consider variability and seasonality, not just the overall averages.

Next, Figures 6 and 7 show that performance differs based on the location and the interaction of multiple operational factors. The geographic map shows that delay pressure isn't evenly distributed. There is a mild east-leaning gradient and higher averages concentrated in the Northeast region. Figure 7 then combines volume, delay, and on-time percentage to show that weaker performance isn't just a function of scale. It shows that several airlines manage similar flight volumes with better outcomes than others. This reinforces the idea that leaders should focus on targeted operational action. The figures point toward closer review of weaker-performing airlines, heightened attention to high-delay months, and deeper investigation into airports where delay pressure is concentrated. Averages alone can mask meaningful operational risk, and effective management requires understanding where, when, and why performance weakens.

Overall, the set of visuals is successful because it progresses in a natural sequence from fundamental baseline comparisons to more sophisticated perspectives of variability, time, geography, and multiple trade-offs. This approach allows management to get a sense not only of which airlines are seemingly better or worse, but also where averages might be masking key variation or concentration. However, the analysis is limited because the data does not reveal the causes of delay and some of the visuals (particularly averages and maps) can exaggerate small differences. Also, without further information, a viewer may mistakenly believe that a one-month rise in delays at an airport or airline necessarily indicates a long-term or location-related problem. Therefore, the visuals should be used as a way to inform decisions and further exploration, but not as a definitive cause-and-effect. In terms of ethics, the report presents patterns by pointing out the lack of context, avoiding over-hype and acknowledging the possibility that automation or default visuals can sometimes exaggerate differences.